

WASP-39b

R-1013

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_260330 · created 2026-07-30T03:10:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461129.896 +/- 0.025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.029 +/- 0.055
Transit depth (Rp/Rs)^2	0.082 +/- 0.31 [%]
Orbital Inclination (inc)	87.51 +/- 1.93
Airmass coefficient 1 (a1)	0.938 +/- 0.023
Airmass coefficient 2 (a2)	0.06 +/- 0.12
Scatter in the residuals of the lightcurve fit is	2.51 %
Best Comparison Star	#3 - [295, 262]
Optimal Method	PSF photometry
Transit Duration (day)	0.08 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.029 ± 0.055 vs 0.1457 (deviation 1.2σ)
Duration fit vs published	0.08 d vs 0.12 d (deviation 0.73σ)
Mid-time O–C	59.7 min
Depth SNR	0.3
Residual scatter	2.51 %

Light curve

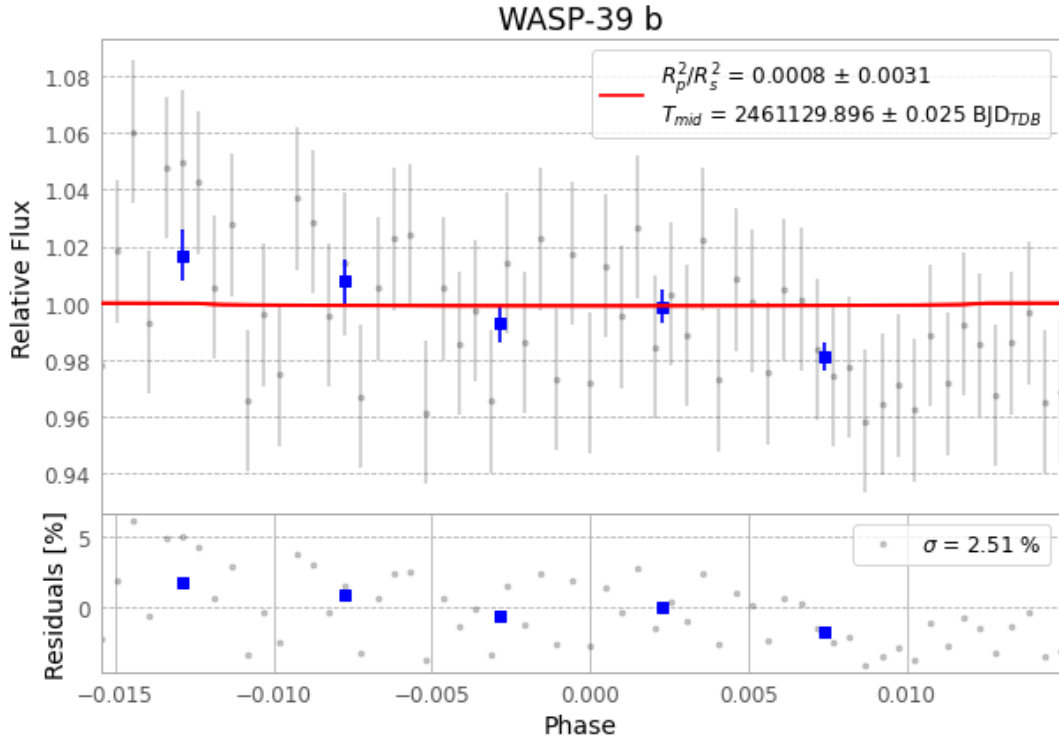


Figure 1 — FinalLightCurve_WASP-39 b_2026-03-30.png (SHA-256 1b861c1cb6279edc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 323], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 84a1a91260242e84...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

60 FITS frames (39 MB), WASP-39260330075115.FITS → WASP-39260330104816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-260330.log (7f77988118b2...), FinalLightCurve_WASP-39 b_2026-03-30.png (1b861c1cb627...), FinalLightCurve_WASP-39 b_2026-03-30.csv (232a45762a02...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 03:10Z.